

Written Test 1 Rewrite

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1. (2a) Show that outer measure is countably subadditive on subsets of $\mathbb{R}$.

Proof. We want to prove that given a sequence $A_1, A_2, \dots$ of subsets of $\mathbb{R}$, then

$$\left| \bigcup_{k=1}^{\infty} A_k \right| \leq \sum_{k=1}^{\infty} |A_k|.$$

First, if $|A_k| = \infty$ for any k , then the RHS is ∞ , and the LHS is ∞ by monotonicity. Otherwise, let $\varepsilon > 0$. Then we can choose a sequence of open intervals $I_{1,k}, I_{2,k}, \dots$ for each A_k such that

$$\sum_{j=1}^{\infty} \ell(I_{j,k}) \leq \frac{\varepsilon}{2^k} + |A_k|.$$

Then

$$\sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \ell(I_{j,k}) \leq \varepsilon + \sum_{k=1}^{\infty} |A_k|.$$

The collection $\{I_{j,k} : j, k \in \mathbb{Z}^+\}$ is countable since it is a countable collection of open intervals, and it covers $\bigcup_{k=1}^{\infty} A_k$, so

$$\left| \bigcup_{k=1}^{\infty} A_k \right| \leq \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \ell(I_{j,k}) \leq \varepsilon + \sum_{k=1}^{\infty} |A_k|.$$

This is true for every $\varepsilon > 0$, so countable subadditivity holds. $\square$

2. (2c) Let (X, S, μ) be a measure space. Let $\{A_n\}_{n=1}^{\infty} \subset S$ such that $A_n \subset A_{n+1}$ for all n . Show that

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Proof. First, if $\mu(A_k) = \infty$ for some k , then the LHS is ∞ by monotonicity, and the RHS is ∞ since $A_j = \infty$ for all $j > k$ by monotonicity. Then assuming $\mu(A_k) < \infty$ for all k and letting $A_0 = \emptyset$ for convenience,

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{j=1}^{\infty} (A_j \setminus A_{j-1}).$$

The RHS is a disjoint union, so

$$\begin{aligned}
\mu\left(\bigcup_{k=1}^{\infty} A_k\right) &= \sum_{j=1}^{\infty} \mu(A_j \setminus A_{j-1}) \\
&= \lim_{N \rightarrow \infty} \sum_{j=1}^N \mu(A_j \setminus A_{j-1}) \\
&= \lim_{N \rightarrow \infty} \sum_{j=1}^N \mu(A_j) - \mu(A_{j-1}) \\
&= \lim_{N \rightarrow \infty} \mu(A_N).
\end{aligned}$$

□

3. (2d) Let $A \subset \mathbb{R}$ and let $F \subset \mathbb{R}$ be a closed set with $A \cap F = \emptyset$. Show that

$$|A \cup F| = |A| + |F|.$$

You may assume that a similar statement has already been proved when an open set G is substituted for F .

Proof. Choose a sequence $I_1, I_2, \dots$ of open intervals whose union $G = \bigcup_{k=1}^{\infty} I_k$ contains $A \cup F$. Then G is open, so $A \subset G \setminus F$ (also open), and thus

$$|A| \leq |G \setminus F|. \quad (1)$$

Then since $G = F \cup (G \setminus F)$,

$$|G| = |F| + |G \setminus F|$$

by the assumed similar statement about open sets.. Adding $|F|$ to (1) yields

$$|A| + |F| \leq |F| + |G \setminus F| = |G| \leq \sum_{k=1}^{\infty} \ell(I_k),$$

so $|A| + |F| \leq |A \cup F|$. The other direction is immediate due to countable subadditivity, so $|A| + |F| = |A \cup F|$. □

4. (3c) Show that outer measure is countably additive on $\mathcal{L}$, the σ -algebra of Lebesgue measurable sets. You may assume that the various equivalent definitions for the term *Lebesgue measurable* have all been proved.

Proof. Given a disjoint sequence $A_1, A_2, \dots$ of sets in $\mathcal{L}$, we can choose a (naturally disjoint) sequence of Borel sets $B_1, B_2, \dots$ where each $B_i \subset A_i$ so that $|A_i \setminus B_i| = 0$. Then $|B_i| \leq |A_i|$ by monotonicity, and countable subadditivity shows that

$$\begin{aligned}
|A_i| &= |(A_i \setminus B_i) \cup B_i| \\
&\leq |A_i \setminus B_i| + |B_i| \\
&= |B_i|,
\end{aligned}$$

so $|A_i| = |B_i|$ for each i . Then

$$\begin{aligned}\left|\bigcup_{k=1}^{\infty} A_k\right| &\geq \left|\bigcup_{k=1}^{\infty} B_k\right| \\ &= \sum_{k=1}^{\infty} |B_k| \\ &= \sum_{k=1}^{\infty} |A_k|.\end{aligned}$$

Since countable subadditivity shows the other direction, the proof is complete. □